

Use hats for estimators

$Cov(aX+bY) = aCov(X,Y) + bCov(Y,X)$
 $Cov(X+a, Y+b) = Cov(X,Y)$
 $Cov(X,X) = Var(X)$
 $Cov(X,Y) = E(XY) - E(X)E(Y)$
 $Cov(X+Y, W+Z) = Cov(X,W) + Cov(Y,W) + Cov(X,Z) + Cov(Y,Z)$
 $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$
 $Corr = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \quad -1 \leq Corr \leq 1$
 $-\sigma_X \sigma_Y \leq Cov \leq \sigma_X \sigma_Y$

640:481 Austin DeCicco Final Sheet

If $X_1, \dots, X_n$ i.i.d. norm r.v.s then
 a. $\bar{X}$ and S are indep
 b. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$
 $Var(S^2) = \frac{2\sigma^4}{n-1}$
 $\int u dv = uv - \int v du$
 $\int u dv dx = u \int v dx - \int u' (\int v dx) dx$

$\mu'_1 = E(X)$
 $\mu'_2 = E(X^2)$
 $\mu'_3 = E(X^3)$
 $\mu'_4 = E(X^4)$
 $\mu'_5 = E(X^5)$
 $\mu'_6 = E(X^6)$
 $\mu'_7 = E(X^7)$
 $\mu'_8 = E(X^8)$
 $\mu'_9 = E(X^9)$
 $\mu'_{10} = E(X^{10})$
 $\mu'_{11} = E(X^{11})$
 $\mu'_{12} = E(X^{12})$
 $\mu'_{13} = E(X^{13})$
 $\mu'_{14} = E(X^{14})$
 $\mu'_{15} = E(X^{15})$
 $\mu'_{16} = E(X^{16})$
 $\mu'_{17} = E(X^{17})$
 $\mu'_{18} = E(X^{18})$
 $\mu'_{19} = E(X^{19})$
 $\mu'_{20} = E(X^{20})$

Gamma: $f(x) = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}$ $\phi_x = \frac{1}{\Gamma(\alpha)} x^{\alpha-1} e^{-x}$ $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy = (\alpha-1)\Gamma(\alpha-1) = (\alpha-1)!$
 $\alpha, \beta, x > 0$ $E(X) = \alpha/\beta$ $Var(X) = \alpha/\beta^2$ $\Gamma(1/2) = \sqrt{\pi}$ $e^x = \sum_{k=0}^\infty \frac{x^k}{k!} = \lim_{n \rightarrow \infty} (1+x/n)^n$ $\chi^2_{\alpha, \nu}$
Exp: $f(x) = \lambda e^{-\lambda x}$ $\phi_x = \lambda e^{-\lambda x}$ if $t < 1/\lambda$ **Chi:** $f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2}$ $\phi_x = \frac{1}{2^{n/2} \Gamma(n/2)}$
 $x, \lambda > 0$ $E(X) = 1/\lambda$ $Var(X) = 1/\lambda^2$ **Special case of gamma** $\alpha = 1/2, \beta = 2$ $x, \nu > 0$ $E(X) = \nu$ $Var(X) = 2\nu$ $\sum_{i=1}^n Z_i^2 = \chi^2_n$ for Z_i i.i.d.
Norm: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$ $\phi_x = e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$ **Geo:** $f(x) = p(1-p)^{x-1}$ $\phi_x = \frac{p}{1-(1-p)e^t}$ $E(X) = 1/p$ $Y(x) = \frac{1-p}{p}$
Bin: $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$ $\phi_x = (pe^t + 1 - p)^n$ $E(X) = np$ $Var(X) = np(1-p)$ **Median:** $n = 2m+1$ $\tilde{X} = Y_{(m+1)}$
Markov: $P(X \geq a) \leq \frac{E(X)}{a}$ $x > 0$ **Chebyshev:** $P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$ $k > 0$ $h(\tilde{x}) = \frac{(2m+1)!}{m!m!} \int_0^{\tilde{x}} f(x) dx \int_{\tilde{x}}^1 f(x) dx$
Order: $g_r(y_r) = \frac{n!}{(r-1)!(n-r)!} \left[\int_{-\infty}^y f(x) dx \right]^{r-1} f(y_r) \left[\int_y^\infty f(x) dx \right]^{n-r}$ $n = 2m$ $\tilde{X} = \frac{1}{2}(Y_m + Y_{m+1})$

If $Y = aX + b$
 $\phi_Y = e^{tb} M_X(at)$
 If X & Y indep
 $Z = X+Y$ $\phi_Z = \phi_X \phi_Y$

$\bar{X} = \sum X_i/n$ $S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$ $E(\bar{X}) = \mu$ $Var(\bar{X}) = \sigma^2/n$ $P(\mu - c \leq \bar{X} \leq \mu + c) \geq 1 - \frac{\sigma^2}{nc^2}$ $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$
Method of moments: r -th sample moment about origin, $M'_r = \sum X_i^r/n$ $M'_1 = \bar{X}$ $M'_2 = \frac{\sum X_i^2}{n}$
 by SL of LN, $M'_r \rightarrow E(X^r)$, thus set $M'_r = \mu'_r$ for estimators.
MLE: $L(\theta) = f(x_1, x_2, \dots, x_n | \theta)$ Take $\partial/\partial \theta$ and set to 0. $\ln(L(\theta))$ might be easier
SL: $\lim_{n \rightarrow \infty} \sum \frac{X_i - \mu}{\sqrt{n}} = N(0,1)$ with prob 1 **CLT:** $\lim_{n \rightarrow \infty} P\left(\frac{\sum (X_i - \mu)}{\sqrt{n}\sigma} \leq a\right) = \Phi(a)$ **WLI:** $P(|\bar{X} - \mu| \geq \epsilon) \rightarrow 0$ as $n \rightarrow \infty$

$H_0: T | H_0 \sim F$
 R 1 X
 A X Z

Unbiased if $E(\hat{\theta}) = \theta$ Asymptotically unbiased if $\lim_{n \rightarrow \infty} E(\hat{\theta}) = \theta$ **CRLB:** $\frac{1}{n} E[(\ln f(x))'^2]$ or $\frac{1}{n} E[\frac{1}{f(x)} \frac{d^2 f(x)}{dx^2}]$
CRLB($\hat{\mu}$) = σ^2/n **CRLB($\hat{\sigma}^2$) = $2\sigma^4/n$** **Critical Region = Rejection Region** **t-score:** $t_{\alpha, \nu}$ **f-score:** f_{α, ν_1, ν_2}
 $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ $f_{\alpha, \nu_1, \nu_2} = f_{1-\alpha, \nu_2, \nu_1}$
 $F = \frac{U/\nu_1}{V/\nu_2} = (\nu_2/\nu_1) (U/V)$ for $U \sim \chi^2_{\nu_1}$, $V \sim \chi^2_{\nu_2}$ $g(t) = \frac{P(\frac{\nu_1 + \nu_2}{2})}{\Gamma(\frac{\nu_1}{2}) \Gamma(\frac{\nu_2}{2})} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} f^{\nu_1/2-1} (1 + \frac{\nu_1}{\nu_2} f)^{-\frac{1}{2}(\nu_1 + \nu_2)}$
 For $F \sim f_{\nu_1, \nu_2}$, $1/F \sim f_{\nu_2, \nu_1}$ $g(t) = \frac{P(\frac{\nu_1 + \nu_2}{2})}{\Gamma(\frac{\nu_1}{2}) \Gamma(\frac{\nu_2}{2})} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} f^{\nu_1/2-1} (1 + \frac{\nu_1}{\nu_2} f)^{-\frac{1}{2}(\nu_1 + \nu_2)}$
Power: $1 - \beta$ $F = \sigma_1^2 S_1^2 / \sigma_2^2 S_2^2 \sim f_{n_1-1, n_2-1}$ $H_0: \mu = 4$ simple, $H_1: \mu < 4$ composite

Difference of means: $Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$ $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$ $T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2}$
Proportions: $Z = \frac{(\sum X_i) - np}{\sqrt{np(1-p)}}$ $Z = \frac{\bar{X} - p}{\sqrt{p(1-p)/n}}$ **Difference of proportions:** $Z = \frac{(\bar{X}_1 - \bar{X}_2) - (p_1 - p_2)}{\sqrt{Var(\bar{X}_1) + Var(\bar{X}_2)}}$
 indep $Var(\bar{X}_i) = \frac{p_i(1-p_i)}{n_i}$
 Estimate σ^2 with $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ Estimate σ_1/σ_2 with $\sigma_1^2 S_1^2 / \sigma_2^2 S_2^2 \sim f_{n_1-1, n_2-1}$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} & \dots \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} & \dots \end{vmatrix}$$

Neymann-Pearson: $H_0 = \theta_0$ and $H_1 = \theta_1$, $L_0 = \prod f(x_i; \theta_0)$, $L_1 = \prod f(x_i; \theta_1)$ $K > 0$

Critical Region C of size α , C is MPCR if $L_0 \leq K$ inside C , $L_0 \geq K$ outside C

Notes: Rename K and to determine test stat then calc K with a known pdf or CI

Likelihood Ratio: $W \cap W^c = \emptyset$, $W \cup W^c = \Omega$ LRT makes no claims about power

$L_W(x_1, \dots, x_n) = \max \prod f(x_i; \theta)$ for $\theta \in W$ $\lambda = \frac{L_W}{L_\Omega}$ Note: For L_Ω use $\hat{\theta}$
 $L_\Omega(x_1, \dots, x_n) = \max \prod f(x_i; \theta)$ for $\theta \in \Omega$ $0 \leq \lambda \leq 1$, $0 < K < 1$

When $H_0 = T$ we expect $\lambda \approx 1$ and when $H_0 = F$ we expect $\lambda \approx 0$.

Thus $\lambda \leq K$ defines the critical region.

P-value: Observed level of significance, lowest α at which H_0 is rejected

Regressions: $\mu_{Y|X} = \int y \cdot w(y|x) dy$ $w(y|x) = f(x, y) / f_X(x)$ $f_X(x) = \int f(x, y) dy$

$\mu_{Y|X} = \alpha + \beta X$ $\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$ $\beta = \sigma_{XY} / \sigma_X^2$ $\alpha = \mu_Y - \beta \mu_X$

$\mu_{Y|X} = \mu_Y + \beta(X - \mu_X)$ $\mu_{X|Y} = \mu_X + \rho \left(\frac{\sigma_X}{\sigma_Y} \right) (Y - \mu_Y)$ Assume $\text{Var}(Y|X) = \text{Var}(Y) = \sigma_Y^2$

X indep var, Y dependent var If Y norm: $w(y|x) = \frac{1}{\sqrt{2\pi}\sigma_Y} e^{-(y - (\alpha + \beta x))^2 / 2\sigma_Y^2}$

$S_{XX} = \sum (x_i - \bar{x})^2 = \overline{(x_i - \bar{x}) \cdot (x_i - \bar{x})}$ $S_{XY} = \sum (x_i - \bar{x})(y_i - \bar{y}) = \overline{(x_i - \bar{x}) \cdot (y_i - \bar{y})}$

$\hat{\beta} = S_{XY} / S_{XX}$ $\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$ $E(\hat{\beta}) = \beta$ $\text{Var}(\hat{\beta}) = \sigma^2 / S_{XX}$

$\hat{\sigma}^2 = (S_{YY} - \hat{\beta} S_{XY}) / n$

Normal Equations for α, β, σ

$\sum (y_i - \alpha - \beta x_i) = 0$

$\sum (y_i - \alpha - \beta x_i) x_i = 0$

$n - \sum (y_i - \alpha - \beta x_i)^2 / \sigma^2 = 0$

$n \hat{\sigma}^2 / \sigma^2 \sim \chi^2_{n-2}$

and indep of $\hat{\beta}$

$\frac{\hat{\beta} - \beta}{\hat{\sigma}} \left(\frac{(n-2) S_{XX}}{n} \right) \sim t_{n-2}$

$\frac{\hat{\beta} - \beta}{\sigma / \sqrt{S_{XX}}} \sim Z$

Transform One Vari Discrete $Y = U(X)$ is one to one, $u'(Y) = X$
 then $f_Y = f_X(u'(y))$ and change domain $\leftarrow$

Otherwise for diffable & monotone $U(X)$ for $f_X(x) \neq 0$

$f_Y = f_X(u'(y)) \cdot |1 / \frac{dy}{dx} u'(y)|$ provided $u'(x) \neq 0$ else $g(y) = 0$

Transform ≥ 2 Var: $f(x_1, x_2)$ given and $Y = U(x_1, x_2)$

Sub expressions for $x_i = \text{func}(Y, x_j)$ obtained from same expression of x_i

Discrete: $g(y, x_2) = f(x_1, x_2) \left| \frac{\partial x_1}{\partial y} \right|$, $g(x_1, y) = f(x_1, x_2) \left| \frac{\partial x_2}{\partial y} \right|$

then integrate out x_j to get g_Y .

$\rightarrow$ Bounds $\leftarrow$

Otherwise: $Y_1 = U_1(x_1, x_2)$, $Y_2 = U_2(x_1, x_2)$ for partial diffable one to one
 $x_1 = W_1(Y_1, Y_2)$, $x_2 = W_2(Y_1, Y_2)$

$g(y_1, y_2) = \int_{x_1, x_2} f(W_1(y_1, y_2), W_2(y_1, y_2)) \cdot |J|$ then integrate for f_{Y_i}